

TARGET-SPECIFIC RECOVERABILITY MAPS FOR REDUCED-LEAD ECG RECONSTRUCTION

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ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature, on *which* lead, a clinician can trust. We give a *target-specific recoverability map*: for a waveform segment s (P/QRS/ST/T), an observed lead set S , and a target lead ℓ , two closed-form numbers derived from one truncated SVD of the segment’s empirical rank-3 spatial (“dipolar”) sub-matrix—an *identifiability* $\eta_{s,\ell}(S)$ (zero iff the target’s low-rank component is recoverable from S) and a *conditioning* $\kappa_{s,\ell}(S)$ (its noise/observed-residual gain). On PTB-XL the map is stable to the truncation tolerance for the well-posed configurations and rank-sensitive (with reported sensitivity) for the ill-posed ones; e.g. it proves *a priori* that precordial ST is not identifiable from limb leads. We add distribution-free calibrated intervals for the empirically predictable residual (conformalized quantile regression with strict train/ calibrate/test fold discipline; within-group coverage near nominal), and contextualize with reconstruction baselines. The map is reconstructor-agnostic and turns “trust the reconstruction” into a per-feature, per-lead, calibrated statement.

Index Terms— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads.

We ask a sharper, reconstructor-independent question: *for a given observed lead set, which target lead’s dipolar morphology is even identifiable, and how well-conditioned is its recovery?* Each waveform segment’s instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis M_s (top-3 singular vectors of the population lead covariance). We stress that M_s is a *population-estimated* (PCA) object: on real PTB-XL it shares

two directions with the classical vectorcardiographic (inverse-Dower) dipole but its third direction differs materially, so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 3).

Contributions. (1) A *target-specific* recoverability map: closed-form per-lead identifiability $\eta_{s,\ell}(S)$ and conditioning $\kappa_{s,\ell}(S)$ from a single truncated SVD, replacing coarse “global rank<3” rules, with a truncation-tolerance sensitivity analysis and record-bootstrap confidence intervals (Sec. ??). (2) A distribution-free calibration layer for the empirically predictable residual, with strict fold discipline and honestly stated within-group marginal coverage (Sec. 2). (3) Evidence on PTB-XL: reconstruction baselines that quantify what each configuration can recover, and a certificate-level safety statement—precordial ST is not identifiable from limb leads—available before any reconstruction (Sec. 3).

Model. Within segment s write the population 12-lead vector as $L_s = \mu_s + M_s d + r_s$, where $M_s \in \mathbb{R}^{12 \times 3}$ is the (population-estimated, PCA) dipolar basis, d the dipole coordinates, and r_s the non-dipolar residual. Observing S gives $y_S = \text{Sel}_S L_s + n$. Let $M_{s,S} = \text{Sel}_S M_s$ have truncated SVD at relative tolerance g ; $M_{s,S}^+$ its truncated pseudo-inverse and $P_{\text{obs}} = M_{s,S}^+ M_{s,S}$ the projector onto the observed dipole coordinate directions.

Theorem 1 (Per-target-lead dipolar identifiability and conditioning). *The mean-centred dipolar estimate $\hat{L}_s = \mu_s + M_s M_{s,S}^+ (y_S - \mu_{s,S})$ satisfies, for each target lead ℓ with e_ℓ the ℓ -th unit vector:*

- (i) *Identifiability. Define $\eta_{s,\ell}(S) = \|e_\ell^\top M_s (I - P_{\text{obs}})\|_2$. If $\eta_{s,\ell}(S) = 0$, the dipolar component of lead ℓ is identifiable from S : $e_\ell^\top M_s d$ is determined by $\text{Sel}_S M_s d$. If $\eta_{s,\ell}(S) > 0$, there is a dipole direction unobserved by S that changes lead ℓ ; its dipolar component is not identifiable at any SNR.*
- (ii) *Conditioning. On the identifiable part, the error from observation noise and from the observed non-dipolar residual is amplified into lead ℓ by $\kappa_{s,\ell}(S) = \|e_\ell^\top M_s M_{s,S}^+\|_2$: $|e_\ell^\top (\hat{L}_s - \mu_s - M_s P_{\text{obs}} d)| \leq \kappa_{s,\ell}(S) (\|\text{Sel}_S r_s\| + \|n\|)$.*

The global $\kappa_s(S) = \|M_s M_{s,S}^+\|_2 = \max_\ell \kappa_{s,\ell}(S)$ is a configuration-level worst case. Both $\eta_{s,\ell}$ and $\kappa_{s,\ell}$ depend

on the estimated M_s and on ϱ ; near-rank-deficient S are reported with an ϱ -sensitivity sweep and record-bootstrap confidence intervals.

The non-dipolar residual (no independence claim on real data). Let $m_s(y_S) = \mathbb{E}[r_s | y_S]$ and $u_s = r_s - m_s(y_S)$, so $\mathbb{E}[u_s | y_S] = \mathbf{0}$. We do *not* assume $u_s \perp y_S$. The Bayes-optimal reconstruction error of the residual is

$$\inf_{\hat{r}} \mathbb{E} \|r_s - \hat{r}(y_S)\|^2 = \mathbb{E} [\text{Var}(r_s | y_S)].$$

Part of r_s (the *empirically predictable residual*) is recovered by a predictor trained on y_S and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—not declared unrecoverable a priori.

Proposition 1 (Strict independence limit – synthetic model only). *If the data are generated so that a component u is statistically independent of y_S (an explicit synthetic construction), then $p(u | y_S) = p(u)$, the Bayes estimate of u is its prior mean, and $\mathbb{E} \|\hat{u} - u\|^2 \geq \text{Var}(u)$ for any estimator. This strict limit is invoked only for the synthetic model; on real ECG we make no such independence claim.*

2. DISTRIBUTION-FREE CALIBRATION

Predictable-residual intervals. For each pre-registered group $g = (S, s, \ell)$ we train a quantile regressor of the target lead’s off-dipole residual on the observed leads (gradient-boosted pinball loss) and apply conformalized quantile regression [3] with a Mondrian (per-group) correction. Fold discipline is strict and leakage-free: basis and quantile model on folds 1–7, conformal calibration on fold 9, a single evaluation on fold 10. We report *within-group marginal coverage* under exchangeability (not per-example conditional coverage), interval width, and group sizes.

Off-dipole flag. We threshold the off-dipole energy at the one-sided $(1 - \alpha)$ conformal quantile over faithful reconstructions; the false-flag rate is $\leq \alpha$ for a fresh reconstruction *exchangeable with the calibration distribution*—not distribution-free under an arbitrary shift.

3. EXPERIMENTS

Data. PTB-XL [4] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [5].

3.1. The recoverability map on PTB-XL

We estimate M_s (rank 3) per segment on the training folds and evaluate the map on 1,200 NORM records (per-lead η, κ with 200-sample bootstrap CIs, truncation tolerance $\varrho=10^{-2}$; Fig. 1). The map is per target lead: a single observed lead (e.g.

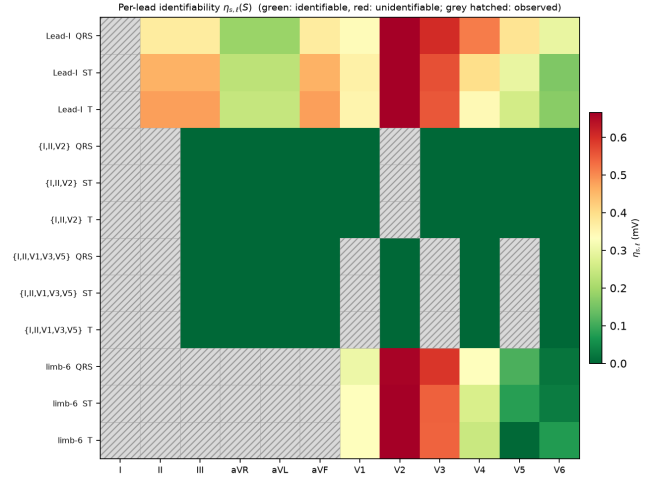


Fig. 1. Per-lead identifiability $\eta_{s,\ell}(S)$ across configurations and segments (green: identifiable, red: unidentifiable; grey hatched: observed). limb-6 leaves all precordial leads unidentifiable.

Lead I) leaves *every* other lead’s dipolar component unidentifiable ($\eta > 0$; rank 1); a dipole-spanning triplet $\{I, II, V2\}$ or the five-lead $\{I, II, V1, V3, V5\}$ makes all targets identifiable ($\eta \approx 0$). For the six limb leads, *all* precordial leads V1–V6 are unidentifiable for QRS/ST/T: the transverse dipole is unobserved, so precordial ST cannot be read from limb leads—a warning available before any reconstruction.

Certificate-level safety. The warning is actionable. On 1,498 delineated fold-10 records, reconstructing precordial ST from limb-6 with a real dipolar reconstructor produces 32.5% fabricated (CI [29.9, 34.9]) and 17.6% masked (CI [15.8, 19.7]) ST-threshold events (a $|ST| > 0.1$ mV crossing present in the reconstruction but not the truth, or vice-versa), at a mean ST error of 0.076 mV—the cost of the certified unidentifiability ($\eta_{ST,\ell} > 0$ for every precordial ℓ). We report ST- threshold events, not diagnoses.

Truncation-tolerance sensitivity. Rank/ κ of well-posed configurations are stable across $\varrho \in \{10^{-4}, \dots, 10^{-1}\}$ (e.g. $\{I, II, V1, V3, V5\}$ is rank 3 throughout), while near-rank-deficient sets are sensitive: on synthetic data $\{V1, V2, V3\}$ is rank 2 at $\varrho=10^{-2}$ but rank 3 with $\kappa \sim 200$ at $\varrho=10^{-4}$. We therefore report a single κ only for well-conditioned configurations and give the ϱ -sweep and bootstrap CIs otherwise; a single 10^4 or 10^5 figure for a degenerate set is not meaningful.

3.2. Calibrated intervals for the predictable residual

For the empirically predictable residual we train a gradient-boosted quantile regressor of each target lead’s off-dipole residual on the observed leads and apply Mondrian CQR per (S, s, ℓ) group, with strict fold discipline (basis and model on

Table 1. Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol (800 test records). “dipolar” = Tier I; “ridge” = per-segment ridge (Tier I+II); “U-Net” = strong arbitrary-mask neural baseline. Best in bold. On limb-6 even the neural baseline cannot recover precordial morphology, as $\eta > 0$ predicts. Numbers auto-generated from `results/fair_baselines.json`.

config	prior-mean	dipolar	ridge	U-Net
{I,II,V2}	0.471	0.234	0.220	0.181
{I,II,V1,V3,V5}	0.484	0.186	0.168	0.164
limb-6	0.484	0.362	0.346	0.239

folds 1–7, calibration on fold 9, a single evaluation on fold 10; nominal $1 - \alpha = 0.90$). Across the pre-registered groups of {I,II,V1,V3,V5} and {I,II,V2}, fold-10 within-group coverage clusters near nominal (0.88–0.95; e.g. {I,II,V1,V3,V5} QRS/V2 0.92, record-bootstrap CI [0.89, 0.95], mean width 0.165 mV), with a few groups slightly below—the expected finite-sample behaviour of within-group marginal coverage. This certifies *what interval* the predictable residual admits, per feature and lead, without any distribution-free claim beyond exchangeability with the calibration fold.

3.3. Baselines and what the map predicts

Table 1 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: per-timepoint waveform RMSE on the same 800 held-out fold-10 records, same segment indices, record-bootstrap CIs. This is the fair comparison—an earlier draft scored the classical baselines on segment-mean vectors and the neural baseline on waveforms, which is not comparable. We include a *strong neural baseline*: an arbitrary-mask conditional 1-D U-Net trained with random lead masks (so one model serves any configuration, no target-lead leakage), the masked-reconstruction architecture the reduced-lead literature uses [1].

On a dipole-spanning set the methods form a monotone ladder—prior-mean $\rightarrow$ dipolar (Tier I) $\rightarrow$ per-segment ridge (Tier I+II) $\rightarrow$ U-Net—confirming there is recoverable non-dipolar content (the calibrated Tier II layer), yet the strong neural model’s margin over a simple ridge is small (QRS 0.164 vs. 0.168 mV on {I,II,V1,V3,V5}). The decisive result is on **limb-6**: *every* reconstructor degrades sharply, and the U-Net—despite its capacity—reaches only QRS 0.239, $1.46\times$ its spanning-set error, still far above what it achieves when the target is identifiable. No amount of model capacity crosses the identifiability ceiling the map’s $\eta > 0$ warning marks out *a priori*: limb $\rightarrow$ precordial morphology is not recoverable, so a low reconstruction error there would be spurious.

Empirical, not physical, dipole. On real PTB-XL the es-

timated M_s shares two directions with the classical inverse-Dower vectorcardiographic space (principal angles 2–6°) but its third direction differs materially (max angle 43–55°, bootstrap CIs excluding small angles). We therefore treat M_s as an *empirical rank-3 spatial subspace*, not a physical cardiac dipole; the earlier synthetic “matches inverse-Dower” check used inverse-Dower to generate the data and is not evidence on real ECG. All certificate quantities are stated with respect to this estimated subspace.

4. CONCLUSION

Reduced-lead ECG reconstruction has a *known low-rank structure per segment*. Exploiting it gives a per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor. It tells a clinician which lead configuration can support a decision, e.g. that precordial ST cannot be read from limb leads.

5. REFERENCES

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